SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Accredited by NAAC A Grade

SIMATS ENGINEERING

CSA47 – DEEP LEARNING

CO2 – ASSESSMENT 3: ASSIGNMENT-2

COURSE OUTCOME COVERED

CO2: *Focuses on various Sequence Modelling and strategies for debugging.*

Annexure A

ASSIGNMENT QUESTIONS

Code of Conduct:

*I **Manoj A / 192421295** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

Student details	
Name	Manoj A
Register No.	192421295
Course	CSA47 – Deep Learning
Assessment	CO2 – Assessment 3: Assignment-2 (Total Marks: 50)
Date	

SIMATS ENGINEERING

Annexure A – Assignment Solutions

Name: Manoj A Register No.: 192421295

Assessment Title: Analyze the Mathematics of a Diabetic Retinopathy Screening Model (Total Marks – 50)

Scenario: "RetinaCare" diabetic retinopathy screening. Every result below is derived by hand from the data items D1–D6 with the formula and the intermediate steps shown; answers are given to four decimal places and information-theoretic quantities use log base 2.

Given data

Item	Data
D1	Feature matrix A = [3 0; 4 5]
D2	$J(w) = \frac{1}{2} w^T H w - b^T w$, $H = [4\ 1; 1\ 4]$, $b = [5; 5]$, $w_0 = [0; 0]$
D3	Classes (No DR, Mild, Moderate, Severe): true p = [0.5, 0.25, 0.125, 0.125]; untrained q_1 = [0.25, 0.25, 0.25, 0.25]; partly trained q_2 = [0.4, 0.3, 0.2, 0.1]
D4	Prevalence = 10%, sensitivity = 90%, specificity = 95%
D5	True score y = 10, noise variance $\sigma^2 = 1$; Model P predictions = [8, 9, 8, 9]; Model Q predictions = [12, 7, 11, 8]
D6	Confusion matrix (100 images): actual referable: 40 predicted referable, 10 predicted non-referable; actual non-referable: 5 predicted referable, 45 predicted non-referable
Design Challenge	Prevalence 2%; a missed referable case costs 10 times an unnecessary referral; sensitivity 90% and specificity 95% unchanged

TASK 1 – Break Down the Audit Problem *[3 Marks]*

The audit follows the model pipeline: representation → loss → optimisation → decision → evaluation. Each data item is a small, self-contained problem in one of these stages.

Item	Mathematical concept	Model stage	Task(s)	Key quantity / question
D1	Singular value decomposition, low-rank approximation (Eckart–Young), energy $\|A\|_F^2 = \Sigma\sigma_i^2$	Representation (feature compression)	2	How much of A's energy is in the dominant direction?
D2	Quadratic form, gradient, Hessian eigen-analysis, convexity, condition number	Loss surface and optimisation	3, 6, 7	Curvature, minimiser w*, speed and stability of GD / momentum
D3	Entropy, cross-entropy, KL divergence (Gibbs inequality)	Loss function (training objective)	4	How far are q_1 and q_2 from the true class distribution p?
D4	Conditional probability, Bayes' rule, likelihood ratios	Decision (what a test result means)	5, 11	P(Referable | Positive / Negative)
D5	Bias–variance–noise decomposition of expected squared error	Evaluation (generalisation of a score model)	8	Which error source dominates Model P and Model Q?
D6	Confusion matrix, accuracy, precision, recall, specificity, F1	Evaluation / decision quality	9, 10	Where does the classifier make mistakes?

Dependencies between items

- **D2 → D3:** D3's cross-entropy is the kind of objective that an optimiser such as D2's gradient descent minimises; D2 is the quadratic model of how that optimisation behaves near a minimum.
- **D1 → D5, D6:** compression of the features limits what any downstream model can learn, so the error behaviour in D5/D6 is conditional on D1's retained 90% of energy.

- **D4 ↔ D6:** D4 states the operating characteristics (sensitivity, specificity) and the population prevalence; D6 is empirical evidence about the same characteristics (recall and specificity measured on 100 images). D5 and D6 both belong to evaluation but concern different outputs (continuous score versus referral decision).
- **D4 + Design Challenge:** prevalence is the only input that changes in Task 11; sensitivity and specificity are carried over unchanged.

Assumptions made

- Logarithms are base 2 for D3; all other results are exact or rounded to four decimals.
- Sensitivity and specificity in D4 are class-conditional properties of the test and do not change with prevalence; results of different patients are independent.
- D2: the Hessian is constant (quadratic loss); gradient descent starts at $w_0 = 0$; momentum uses $v_{(\square+1)} = \beta v_{\square} - \eta \nabla J(w_{\square})$, $w_{(\square+1)} = w_{\square} + v_{(\square+1)}$ with $v_0 = 0$.
- D5: the four predictions of each model are treated as the full set of retrained models, so variance is the population variance (÷ 4); the sample variance (÷ 3) is also reported as a check.
- D6 is a representative sample for sensitivity and specificity, but its prevalence (50%) is not the deployment prevalence.

TASK 2 – Examine the Compressibility of the Feature Matrix *[6 Marks]*

Step 1 – Form A^TA

$$A^TA = \begin{vmatrix} 3 & 4 \\ 0 & 5 \end{vmatrix} \begin{vmatrix} 3 & 0 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} 25 & 20 \\ 20 & 25 \end{vmatrix}$$

Step 2 – Eigenvalues of A^TA (squares of the singular values)

$$\det(A^TA - \lambda I) = (25 - \lambda)^2 - 20^2 = 0 \;\Rightarrow\; 25 - \lambda = \pm 20 \;\Rightarrow\; \lambda_1 = 45,\ \lambda_2 = 5$$

Check: trace = 25 + 25 = 50 = 45 + 5; determinant = 625 − 400 = 225 = 45 × 5.

Step 3 – Singular values

$$\sigma_1 = \sqrt{45} = 6.7082, \quad \sigma_2 = \sqrt{5} = 2.2361, \quad \sigma_1/\sigma_2 = 3.0000$$

Check: det A = 15 = $\sigma_1\sigma_2 = \sqrt{225}$. The ratio $\sigma_1/\sigma_2 = 3$ is also the condition number $\kappa(A)$ of the matrix.

Step 4 – Right singular vectors V (eigenvectors of A^TA)

$\lambda_1 = 45$: $(A^TA - 45I)v = [-20\ 20;\ 20\ -20]v = 0 \Rightarrow v_1 = (1, 1)/\sqrt{2} = (0.7071, 0.7071)$.

$\lambda_2 = 5$: $(A^TA - 5I)v = [20\ 20;\ 20\ 20]v = 0 \Rightarrow v_2 = (1, -1)/\sqrt{2} = (0.7071, -0.7071)$. Note $v_1 \cdot v_2 = 0$.

Step 5 – Left singular vectors $U = AV\,\Sigma^{-1}$ ($u_i = Av_i / \sigma_i$)

$Av_1 = (3, 9)/\sqrt{2} \Rightarrow u_1 = (3, 9)/(\sqrt{2}\cdot\sqrt{45}) = (1, 3)/\sqrt{10} = (0.3162, 0.9487)$.

$Av_2 = (3, -1)/\sqrt{2} \Rightarrow u_2 = (3, -1)/(\sqrt{2}\cdot\sqrt{5}) = (3, -1)/\sqrt{10} = (0.9487, -0.3162)$. Note $u_1 \cdot u_2 = (3 - 3)/10 = 0$.

Step 6 – The decomposition $A = U\Sigma V^T$

$$A = \begin{vmatrix} 1/\sqrt{10} & 3/\sqrt{10} \\ 3/\sqrt{10} & -1/\sqrt{10} \end{vmatrix} \begin{vmatrix} \sqrt{45} & 0 \\ 0 & \sqrt{5} \end{vmatrix} \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} = \begin{vmatrix} 3 & 0 \\ 4 & 5 \end{vmatrix}$$

Verification of row 1: $(1/\sqrt{20})\cdot[\sqrt{45} + 3\sqrt{5},\ \sqrt{45} - 3\sqrt{5}] = (1/\sqrt{20})\cdot[6\sqrt{5}, 0] = [3, 0]$; row 2: $(1/\sqrt{20})\cdot[8\sqrt{5}, 10\sqrt{5}] = [4, 5]$. So $U\Sigma V^T$ reproduces A exactly.

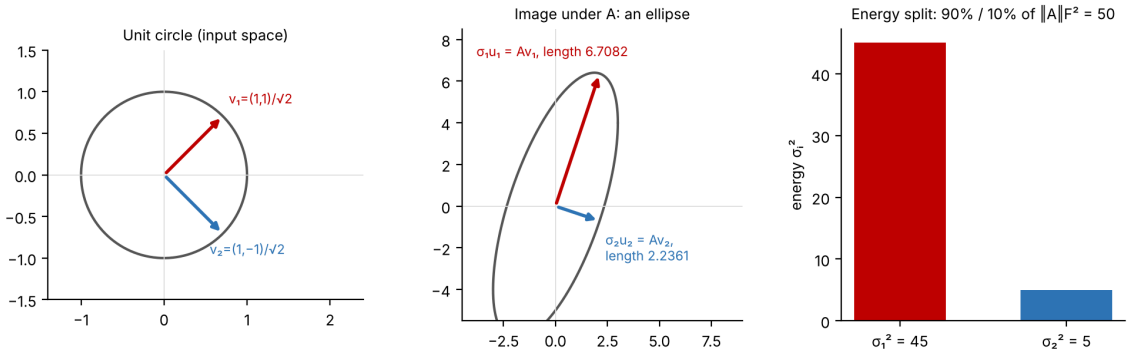


Figure 1: A maps the unit circle to an ellipse whose semi-axes are $\sigma_1 = 6.7082$ and $\sigma_2 = 2.2361$ (left, middle); the energy split is 45 : 5 (right).

Step 7 – Rank-1 approximation and its error

$$A_1 = \sigma_1 u_1 v_1^T = \sqrt{45} \cdot (1,3)^T/\sqrt{10} \cdot (1,1)/\sqrt{2} = (\sqrt{45}/\sqrt{20}) \cdot [1\ 1;\ 3\ 3] = 1.5 \cdot [1\ 1;\ 3\ 3]$$

$$A_1 = \begin{vmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{vmatrix}, \qquad A - A_1 = \begin{vmatrix} 1.5 & -1.5 \\ -0.5 & 0.5 \end{vmatrix}$$

$$\|A - A_1\|_F = \sqrt{(1.5^2 + 1.5^2 + 0.5^2 + 0.5^2)} = \sqrt{5} = 2.2361 = \sigma_2$$

This equals σ_2, as the Eckart–Young theorem predicts: the best rank-1 approximation leaves exactly the discarded singular value as its Frobenius (and spectral) error.

Step 8 – Energy retained

$$\|A\|_F^2 = 3^2 + 0^2 + 4^2 + 5^2 = 50 = \sigma_1^2 + \sigma_2^2 = 45 + 5$$

$$\text{Energy retained by rank 1} = \sigma_1^2 / (\sigma_1^2 + \sigma_2^2) = 45/50 = 0.9000 = 90.00\% \qquad (\text{lost: } 10.00\%)$$

$$\text{Relative Frobenius error} = \|A - A_1\|_F / \|A\|_F = \sqrt{(5/50)} = 0.3162\ (31.62\%)$$

Interpretation and verdict

- **Ratio:** $\sigma_1/\sigma_2 = 3$ means the dominant direction is three times as strong in amplitude and nine times in energy (45 : 5); the matrix is strongly, but not overwhelmingly, dominated by one direction. The dominant input direction $v_1 = (1,1)/\sqrt{2}$ weights the two features equally.
- **What A_1 keeps and loses:** A_1 has identical columns, so it preserves the "overall brightness" pattern (row 2 is about three times row 1) but cannot reproduce the zero in A's first row: A has [3, 0] and A_1 has [1.5, 1.5], so individual entries are off by up to 1.5.
- **Justified?** By the energy criterion (≥ 90% retained) rank-1 is acceptable, and the 31.62% relative error is the price. However, for this 2×2 matrix there is no storage saving (rank 1 needs σ_1, u_1, $v_1 = 5$ numbers versus 4 entries), so the justification is conceptual: in the real, much larger feature matrix the same test (energy retained versus downstream accuracy) should be applied before compressing.

TASK 3 – Analyze the Curvature of the Loss Surface *[5 Marks]*

$$J(w) = \tfrac{1}{2}\, w^T H w - b^T w, \qquad \nabla J(w) = Hw - b, \qquad \nabla^2 J(w) = H \ \text{(constant)}$$

Eigenvalues and eigenvectors of H = [4 1; 1 4]

$$\det(H - \lambda I) = (4 - \lambda)^2 - 1 = 0 \ \Rightarrow\ \lambda = 4 \pm 1: \ \lambda_1 = 5,\ \lambda_2 = 3$$

$\lambda_1 = 5$: $(H - 5I)v = [-1\ 1;\ 1\ -1]v = 0 \ \Rightarrow\ v_1 = (1, 1)/\sqrt{2}$. $\lambda_2 = 3$: $(H - 3I)v = [1\ 1;\ 1\ 1]v = 0 \ \Rightarrow\ v_2 = (1, -1)/\sqrt{2}$.

Curvature directions: J rises most steeply along (1, 1) (curvature 5, high) and most gently along (1, −1) (curvature 3, low). The level sets are ellipses whose axes are in the ratio $\sqrt{(5/3)} = 1.2910$ — nearly circular.

Positive definiteness, convexity, uniqueness

Both eigenvalues are positive (5 and 3); equivalently the leading minors are $4 > 0$ and $\det H = 15 > 0$. Hence H is **positive definite**: $x^T Hx \geq 3\|x\|^2 > 0$ for $x \neq 0$. Because the Hessian is positive definite everywhere, J is **strictly convex**, has no saddle points or local minima other than the global one, and its minimiser is **unique**.

Condition number and gradient descent

$$\kappa(H) = \lambda_{max} / \lambda_{min} = 5 / 3 = 1.6667$$

A condition number close to 1 means the curvature is almost the same in every direction, so one step size suits both directions and GD does not zig-zag. The error contracts by $|1 - \eta\lambda_i|$ per step in each eigen-direction; the best fixed step is $\eta = 2/(\lambda_{max} + \lambda_{min}) = 0.25$, giving the rate $(\kappa - 1)/(\kappa + 1) = 0.2500$ per step. Large κ would force $\eta < 2/\lambda_{max}$ and make the λ_{min} direction crawl.

Minimiser $w^* = H^{-1}b$ and verification

$$H^{-1} = (1/15)\cdot[4\ -1;\ -1\ 4], \quad w^* = H^{-1}b = (1/15)\cdot(4\cdot 5 - 5, -5 + 4\cdot 5) = (1/15)\cdot(15, 15) = (1, 1)$$

$$\nabla J(w^*) = Hw^* - b = (5, 5) - (5, 5) = (0, 0)\ \checkmark \qquad J(w^*) = \tfrac{1}{2}\cdot(1,1)\cdot(5,5) - (5,5)\cdot(1,1) = 5 - 10 = -5$$

Note on the wording "J(w) = 0": the quantity that equals zero at the minimiser is the gradient $\nabla J(w^*) = 0$, which is what verifies $w^* = (1, 1)$. The loss value itself is $J(w^*) = -5$ (the minimum value, equal to $-\tfrac{1}{2}b^T H^{-1}b$); $J(w_0) = 0$ at the starting point. Also $J(w) = -5 + \tfrac{1}{2}(w - w^*)^T H(w - w^*)$, which gives $J(0) = -5 + \tfrac{1}{2}\cdot 10 = 0$ ✓.*

TASK 4 – Differentiate the Models using Information Theory *[5 Marks]*

Formulas: $H(p) = -\Sigma\ p_i \log_2 p_i$; $H(p, q) = -\Sigma\ p_i \log_2 q_i$; $KL(p\|q) = \Sigma\ p_i \log_2(p_i/q_i)$; $H(p, q) = H(p) + KL(p\|q)$.

Entropies and cross-entropy

$$H(p) = 0.5\cdot 1 + 0.25\cdot 2 + 0.125\cdot 3 + 0.125\cdot 3 = 0.5 + 0.5 + 0.375 + 0.375 = 1.7500 \text{ bits}$$

$$H(q_1) = 4 \times (0.25\cdot 2) = 2.0000 \text{ bits} \quad (\text{uniform over 4 classes} = \log_2 4)$$

$$H(p, q_1) = -\Sigma\ p_i \log_2 0.25 = 2\cdot\Sigma\ p_i = 2.0000 \text{ bits}$$

Decomposition $H(p, q_1) = H(p) + KL(p\|q_1)$ verified numerically

$$KL(p\|q_1) = 0.5\cdot\log_2 2 + 0.25\cdot\log_2 1 + 0.125\cdot\log_2 0.5 + 0.125\cdot\log_2 0.5 = 0.5 + 0 - 0.125 - 0.125 = 0.2500 \text{ bits}$$

$$H(p) + KL(p\|q_1) = 1.7500 + 0.2500 = 2.0000 = H(p, q_1)\ \checkmark$$

Class-wise comparison of the two models

Class	p	q_1	q_2	$-p\cdot\log_2 q_1$	$-p\cdot\log_2 q_2$	$p\cdot\log_2(p/q_1)$	$p\cdot\log_2(p/q_2)$
No DR	0.5000	0.2500	0.4000	1.0000	0.6610	+0.5000	+0.1610
Mild	0.2500	0.2500	0.3000	0.5000	0.4342	0.0000	−0.0658
Moderate	0.1250	0.2500	0.2000	0.2500	0.2902	−0.1250	−0.0848
Severe	0.1250	0.2500	0.1000	0.2500	0.4152	−0.1250	+0.0402
Total	**1.0000**	**1.0000**	**1.0000**	**$H(p,q_1)$ = 2.0000**	**$H(p,q_2)$ = 1.8007**	**$KL(p\|q_1)$ = 0.2500**	**$KL(p\|q_2)$ = 0.0507**

Check for q_2: $H(p) + KL(p\|q_2) = 1.7500 + 0.0507 = 1.8007 = H(p, q_2)$ ✓.

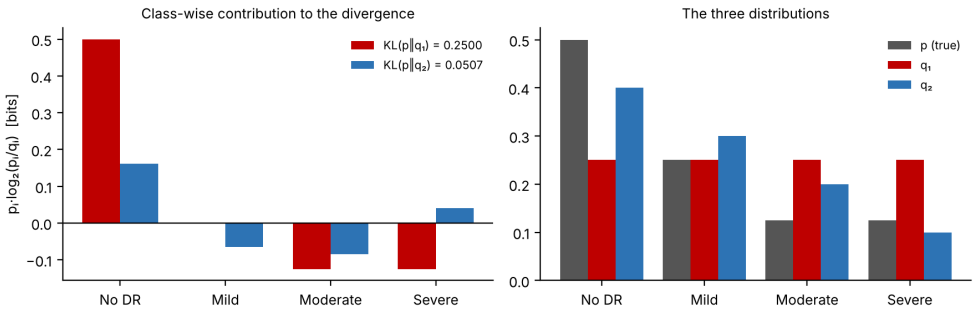


Figure 2: class-wise contributions $p_i \cdot \log_2(p_i/q_i)$ (left) and the three distributions (right).

- **How the divergences differ:** $KL(p\|q_1) = 0.2500$ bits versus $KL(p\|q_2) = 0.0507$ bits, so the partly trained model is about 4.93 times closer to the truth; its cross-entropy falls from 2.0000 to 1.8007 bits, only 0.0507 bits above the floor $H(p) = 1.7500$.
- **Which classes contribute:** for q_1 the divergence comes almost entirely from **No DR** (+0.5000, because q_1 gives it 0.25 instead of 0.5), partly cancelled by Moderate and Severe (−0.1250 each, where q_1 gives too much mass). For q_2 the largest positive term is again **No DR** (+0.1610, $q = 0.4 < 0.5$), followed by **Severe** (+0.0402, $q = 0.10 < 0.125$); Mild and Moderate are over-weighted and contribute negatively. Individual terms may be negative; only their sum is guaranteed ≥ 0 (Gibbs' inequality). Clinically, q_2 still under-weights the rarest and most dangerous class (Severe).
- **Zero divergence:** $KL(p\|q) = 0$ if and only if $q = p$ for every class with $p > 0$. The cross-entropy then equals its minimum $H(p) = 1.7500$ bits: no "extra bits" are wasted by using the model instead of the true distribution. (It certifies the class-probability estimates, not per-image correctness.)

TASK 5 – Analyze the Screening Decision using Bayes' Rule *[4 Marks]*

Let R = referable retinopathy, + = positive screen. $P(R) = 0.10$, $P(+ \mid R) = 0.90$ (sensitivity), $P(- \mid \neg R) = 0.95$ (specificity), so $P(+ \mid \neg R) = 0.05$. Probability tree for 1,000 screened patients (Figure 3, left):

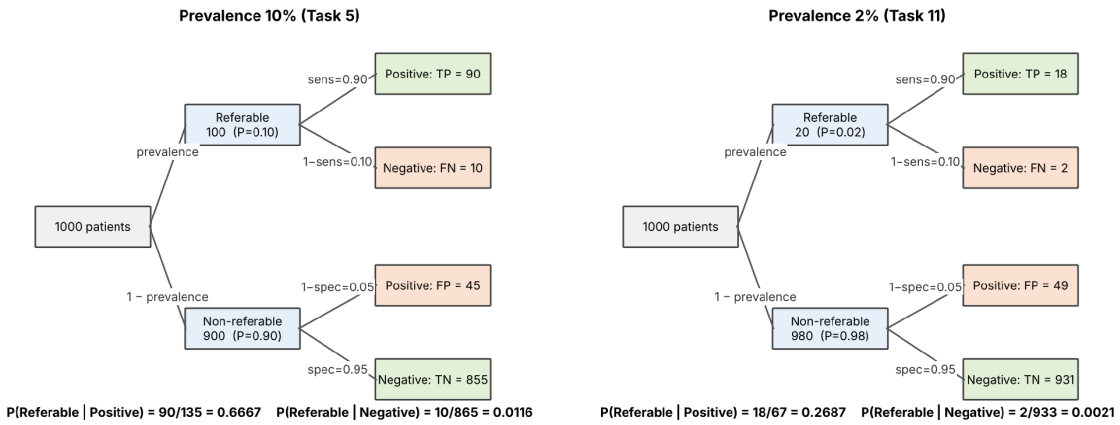


Figure 3: probability trees for prevalence 10% (Task 5) and 2% (Task 11).

$$P(+) = P(+|R)P(R) + P(+|\neg R)P(\neg R) = 0.9 \cdot 0.10 + 0.05 \cdot 0.90 = 0.09 + 0.045 = 0.1350$$

$$P(R \mid +) = 0.09 / 0.135 = 0.6667$$

$$P(-) = 0.10 \cdot 0.10 + 0.95 \cdot 0.90 = 0.01 + 0.855 = 0.8650; \quad P(R \mid -) = 0.01 / 0.865 = 0.0116$$

Counts per 1,000: TP = 90, FN = 10, FP = 45, TN = 855; positives = 135, of which 90 are truly referable. Likelihood-ratio check: $LR^+ = 0.90/0.05 = 18$, prior odds = 0.10/0.90 = 0.1111, posterior odds = 18 × 0.1111 = 2.0 ⇒ P(R | +) = 2/3 = 0.6667 ✓.

Why P(R | +) is much lower than the sensitivity

Sensitivity answers "given disease, how often is the test positive?" (0.90), while P(R | +) answers the reverse question "given a positive, how often is there disease?" (0.6667). They differ because of the base rate: the non-referable group is nine times larger than the referable group, so even a 5% false-positive rate generates 45 false positives, half as many as the 90 true positives. A positive result raises the probability of disease from 10% to 66.67% — a gain of $LR^+ = 18$ on the odds — but does not make it certain.

Is a positive result alone sufficient grounds for treatment?

No. One in three positives (33.33%) is a false alarm, so a positive screen is grounds for **confirmatory examination and referral**, not for treatment. A negative result is reassuring but not zero-risk: P(R | −) = 1.16%, i.e. about 10 missed cases among the 865 screened-negative patients per 1,000 screened.

TASK 6 – Trace the Convergence of Gradient Descent *[5 Marks]*

Update: $w_{(k+1)} = w_k - \eta \nabla J(w_k)$, $\nabla J(w) = Hw - b$, $\eta = 0.1$, $w_0 = (0, 0)$, $w^* = (1, 1)$, $J(w^*) = -5$.

k	w_k	$\nabla J(w_k) = Hw_k - b$	$J(w_k)$	$\|w_k - w^*\|$	ratio to previous
0	(0, 0)	(−5, −5)	0.0000	1.4142	–
1	(0.5, 0.5)	(−2.5, −2.5)	−3.7500	0.7071	0.5000
2	(0.75, 0.75)	(−1.25, −1.25)	−4.6875	0.3536	0.5000
3	(0.875, 0.875)	(−0.625, −0.625)	−4.9219	0.1768	0.5000

Worked step 1: $\nabla J(w_0) = H \cdot 0 - b = (-5, -5)$, so $w_1 = (0, 0) - 0.1 \cdot (-5, -5) = (0.5, 0.5)$; $J(w_1) = \tfrac{1}{2} \cdot (0.5, 0.5) \cdot (2.5, 2.5) - (5, 5) \cdot (0.5, 0.5) = 1.25 - 5 = -3.7500$. Step 2: $Hw_1 = (2.5, 2.5)$, $\nabla J = (-2.5, -2.5)$, $w_2 = (0.75, 0.75)$, $J(w_2) = 2.8125 - 7.5 = -4.6875$. Step 3: $\nabla J(w_2) = (-1.25, -1.25)$, $w_3 = (0.875, 0.875)$, $J(w_3) = 3.8281 - 8.75 = -4.9219$.

Pattern and its link to the eigenvalues of H

$$e_k = w_k - w^*, \quad e_{k+1} = (I - \eta H)\, e_k, \quad \text{eigenvalues of } (I - \eta H): \ 1 - 0.1 \cdot 5 = 0.5 \ \text{ and } \ 1 - 0.1 \cdot 3 = 0.7$$

- The distance to w^* **halves each step** (1.4142 → 0.7071 → 0.3536 → 0.1768): linear (geometric) convergence with ratio exactly 0.5000.
- Here $e_0 = (-1, -1)$ lies entirely along $v_1 = (1, 1)/\sqrt{2}$ (the $\lambda = 5$ direction), because $b \propto (1, 1)$ and $w_0 = 0$. Only the factor $1 - \eta\lambda_1 = 0.5$ is active, hence $\|e_k\| = 1.4142 \cdot 0.5^k$.
- The loss gap $J - J^* = \tfrac{1}{2} e^T H e$ shrinks by $0.5^2 =$ **0.25** per step (5 → 1.25 → 0.3125 → 0.0781), matching the observed J values: $-5 + \{5, 1.25, 0.3125, 0.0781\}$.
- From a generic start the slower factor 0.7 (the $\lambda = 3$ direction) would dominate in the long run: the worst-case rate is $\max|1 - \eta\lambda_i| = 0.7$; it can be improved to 0.25 with $\eta = 0.25$.

TASK 7 – Differentiate Momentum from Plain Gradient Descent *[4 Marks]*

Momentum (heavy-ball) update with $\beta = 0.9$, $\eta = 0.1$, $v_0 = 0$: $v_{k+1} = \beta v_k - \eta \nabla J(w_k)$, $w_{k+1} = w_k + v_{k+1}$.

k	Momentum: v_k	Momentum: w_k	$J(w_k)$	$\|w_k - w^*\|$	Plain GD: w_k	$J(w_k)$	$\|w_k - w^*\|$

1	(0.5, 0.5)	(0.5, 0.5)	−3.7500	0.7071	(0.5, 0.5)	−3.7500	0.7071
2	(0.7, 0.7)	(1.2, 1.2)	−4.8000	0.2828	(0.75, 0.75)	−4.6875	0.3536
3*	(0.53, 0.53)	(1.73, 1.73)	−2.3355	1.0324	(0.875, 0.875)	−4.9219	0.1768
4*	(0.112, 0.112)	(1.842, 1.842)	−1.4552	1.1908	(0.9375, 0.9375)	−4.9805	0.0884

Worked: k = 1: $\nabla J(w_0) = (-5, -5)$, $v_1 = 0.9 \cdot 0 + 0.1 \cdot 5 = 0.5$, $w_1 = 0.5$. k = 2: $\nabla J(w_1) = (-2.5, -2.5)$, $v_2 = 0.9 \cdot 0.5 + 0.25 = 0.70$, $w_2 = 0.5 + 0.70 = 1.20$. (*) Rows 3–4 are extra iterations to show what happens next: $\nabla J(w_2) = (1, 1)$, $v_3 = 0.9 \cdot 0.7 - 0.1 = 0.53$, $w_3 = 1.73$.

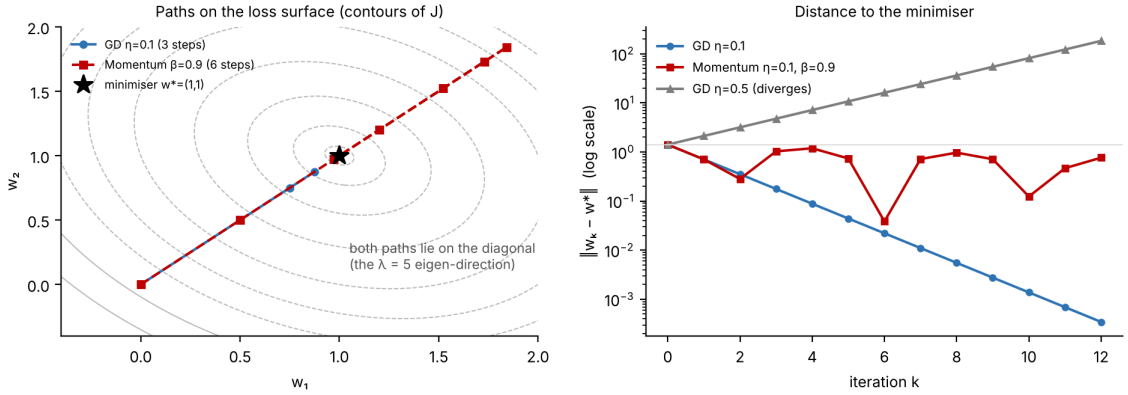


Figure 4: left – GD and momentum paths (both on the diagonal); right – distance to w for GD η = 0.1, momentum, and the divergent η = 0.5.*

Comparison, overshoot and its cause

- After two iterations momentum is **closer** to w* (0.2828 versus 0.3536; J = −4.8000 versus −4.6875) because the accumulated velocity gives larger steps.
- **It overshoots:** $w_2 = (1.2, 1.2)$ is already past w* = (1, 1), and $w_3 = (1.73, 1.73)$ is further past it (distance 1.0324, larger than the 0.7071 reached after the first step). The cause is the velocity memory: $v_2 = 0.70$ carries the iterate through w* while the gradient (now pointing backwards) is still small, and with β = 0.9 the old velocity is damped by only 10% per step.
- Modal analysis: in an eigen-direction with curvature λ the iterate obeys $z^2 - (1 + \beta - \eta\lambda)z + \beta = 0$. For λ = 5: $z^2 - 1.4z + 0.9 = 0$, discriminant 1.96 − 3.60 < 0, complex roots of modulus $\sqrt{\beta} = 0.9487$ (λ = 3 gives the same modulus). The motion is **under-damped**: it oscillates around w* and decays only like 0.9487^k, much slower than plain GD's 0.5^k. For this well-conditioned problem (κ = 1.6667) momentum is therefore worse than plain GD in the long run; its advantage appears for large κ.

Learning-rate bound for plain gradient descent

$$|1 - \eta\lambda_i| < 1 \text{ for all } i \iff 0 < \eta < 2/\lambda_{max} = 2/5 = 0.4000$$

(For heavy-ball momentum the corresponding condition is $0 < \eta < 2(1 + \beta)/\lambda_{max} = 0.76$.) η = 0.1 is well inside the bound; η = 0.4 would give factor −1 along v_1 (endless oscillation).

Why η = 0.5 fails

Along v_1 the error factor is $1 - 0.5 \cdot 5 = \mathbf{-1.5}$, so |factor| > 1: each step overshoots the minimiser by 1.5 times the current error and flips its sign.

k	w□	∇J(w□)	J(w□)	‖w□ − w*‖
0	(0, 0)	(−5, −5)	0.0000	1.4142
1	(2.5, 2.5)	(7.5, 7.5)	6.2500	2.1213
2	(−1.25, −1.25)	(−11.25, −11.25)	20.3125	3.1820

3	(4.375, 4.375)	(16.875, 16.875)	51.9531	4.7730

Trace: $w_1 = 0 - 0.5\cdot(-5) = 2.5$; $\nabla J(w_1) = 5\cdot 2.5 - 5 = 7.5$; $w_2 = 2.5 - 0.5\cdot 7.5 = -1.25$; $\nabla J(w_2) = -11.25$; $w_3 = -1.25 + 5.625 = 4.375$. The error $e_{\square} = (-1.5)^k e_0$ grows geometrically with alternating sign (−1, +1.5, −2.25, +3.375 per coordinate) and the loss increases without bound: GD diverges.

TASK 8 – Decompose the Prediction Error *[4 Marks]*

Expected squared error = Bias² + Variance + σ², Bias = $E[\hat{f}] - y$, Variance = $E[(\hat{f} - E[\hat{f}])^2]$, $y = 10$, $\sigma^2 = 1$

	Model P [8, 9, 8, 9]	**Model Q [12, 7, 11, 8]**
Mean prediction $E[\hat{f}]$	(8+9+8+9)/4 = 8.5	(12+7+11+8)/4 = 9.5
Bias = $E[\hat{f}] - y$	8.5 − 10 = −1.5	9.5 − 10 = −0.5
Bias²	2.2500	0.2500
Deviations from the mean	−0.5, +0.5, −0.5, +0.5	+2.5, −2.5, +1.5, −1.5
Variance (÷ 4)	(4 × 0.25)/4 = 0.2500	(6.25 + 6.25 + 2.25 + 2.25)/4 = 4.2500
Noise σ²	1.0000	1.0000
Expected error	**2.25 + 0.25 + 1 = 3.5000**	**0.25 + 4.25 + 1 = 5.5000**
Check: mean $(\hat{f} - y)^2$ + σ²	(4+1+4+1)/4 + 1 = 3.5000 ✓	(4+9+1+4)/4 + 1 = 5.5000 ✓

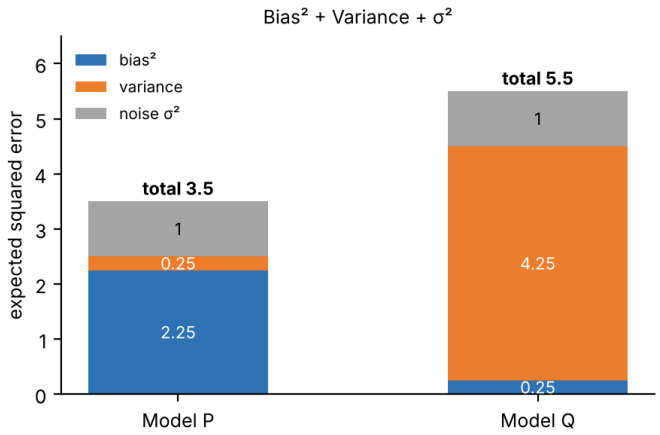


Figure 5: error decomposition of Models P and Q.

- **Dominant component:** Model P is dominated by **bias²** (2.25 of 3.50 total; 90% of its reducible error) – it systematically under-predicts by 1.5 (under-fitting / miscalibration) but its predictions are tightly clustered. Model Q is dominated by **variance** (4.25 of 5.50) – it is nearly unbiased on average (−0.5) but individual predictions scatter by ±1.5 to ±2.5 (over-sensitivity to the training run / over-fitting).
- **Preferable model:** Model P (3.5000 versus 5.5000). Its error is mostly a fixed offset that can be corrected by recalibration or a richer model, whereas Q's variance means any single deployed instance can be far off. (With the sample variance ÷ 3 the totals are 3.5833 and 6.9167, the same ranking.) Both share the irreducible noise floor σ² = 1; the reducible error is 2.5 for P and 4.5 for Q.

TASK 9 – Examine the Error Profile of the Classifier *[3 Marks]*

From D6: TP = 40, FN = 10, FP = 5, TN = 45, N = 100 (50 actual referable, 50 actual non-referable).

Metric	**Formula**	**Value**
Accuracy	(TP + TN)/N = 85/100	0.8500

Error rate	(FP + FN)/N = 15/100	0.1500
Precision	TP/(TP + FP) = 40/45	0.8889
Recall (sensitivity)	TP/(TP + FN) = 40/50	0.8000
Specificity	TN/(TN + FP) = 45/50	0.9000
F1-score	2PR/(P + R) = 2·0.8889·0.8/1.6889 = 2TP/(2TP + FP + FN) = 80/95	0.8421

- **More frequent error:** false negatives (10) are twice as frequent as false positives (5). A false negative is a referable patient told "non-referable": untreated retinopathy can progress to vision loss, an irreversible harm. A false positive costs an unnecessary specialist visit.
- **Why accuracy alone is incomplete:** it weights both error types equally and hides the 10 : 5 split, and it depends on prevalence. This test set is 50% referable; at the 10% deployment prevalence, a useless classifier that labels everybody "non-referable" scores 90% accuracy – higher than 85% – while detecting nothing (recall 0). Recall, specificity and the cost of each error must be reported alongside accuracy.

TASK 10 – Relate the Findings to the Audit Question *[3 Marks]*

Audit question: is the model's compression, loss, optimiser and error behaviour sound enough for clinical use?

Finding (computed result)	Supports soundness	Raises a concern
Task 2: rank-1 keeps 90.00% of the energy; $\sigma_1/\sigma_2 = 3$	Compression keeps the dominant structure	Relative error 31.62%; no saving on a 2×2 matrix – needs validation at full scale
Task 3: H positive definite, $\kappa = 1.6667$, unique $w^* = (1, 1)$	Convex, well-conditioned loss: the optimiser has one well-defined target	–
Task 4: KL falls from 0.2500 to 0.0507 bits; $H(p,q_2) = 1.8007$ vs $H(p) = 1.7500$	Training moves the model towards the true class distribution	q_2 still under-weights Severe (0.10 vs 0.125)
Tasks 6–7: GD ratio 0.5 per step; stable for $\eta < 0.4$; momentum $\beta = 0.9$ overshoots ($w_3 = 1.73$)	Plain GD with $\eta = 0.1$ converges reliably	$\beta = 0.9$ is under-damped; $\eta \geq 0.4$ would diverge – settings must stay inside the bound
Task 8: P = 3.5000, Q = 5.5000; P biased by −1.5	P is stable (variance 0.25)	P under-predicts severity by 1.5 – systematic under-grading could translate into missed cases; Q is erratic (variance 4.25)
Tasks 5, 9: P(R|+) = 0.6667, P(R|−) = 0.0116; recall 0.8000, specificity 0.9000, FN = 10 > FP = 5	NPV = 98.84% at 10% prevalence; precision 0.8889	Recall 0.8000 is below the 0.90 claimed in D4 (95% Wilson interval for 40/50: 0.6696–0.8876); 1 in 3 positives is false

- **Missed referable cases:** the miss rate is 1 − sensitivity. With the D4 value (0.90) there are 10 missed cases per 1,000 patients at 10% prevalence; with the D6 estimate (0.80) there would be 20 per 1,000, and the posterior risk after a negative screen would rise from 1.16% to 2.41%. The error profile (FN = 2 × FP) is where the concern lies, not in the compression, loss or optimiser.
- **Conclusion traced to the numbers:** the mathematical design choices are sound – energy 90.00%, convex loss with $\kappa = 1.6667$, KL 0.2500 → 0.0507, linear convergence for $\eta < 0.4$ – but approval for clinical use should be **conditional** on re-estimating sensitivity and specificity on a larger sample, because the measured recall (0.8000) is lower than the claimed 0.90 and drives the missed-case risk.

TASK 11 – Design Challenge: Analyze the Impact of Low Prevalence *[8 Marks]*

Changed requirement: prevalence 2% (was 10%); a missed referable case costs 10 units and an unnecessary referral 1 unit; sensitivity 90% and specificity 95% unchanged.

A. Impact analysis (2 marks)

Depends on prevalence	Does not depend on prevalence
P(R | +) and P(R | −) (Task 5), P(+) Accuracy, error rate, precision, F1 (Task 9: they mix both classes in the proportions of the sample) Counts of FP and FN per 1,000, and the expected cost KL / cross-entropy of D3 (they depend on the class distribution p, which would shift with prevalence)	Sensitivity (0.90), specificity (0.95), recall, and the likelihood ratios $LR^{+} = 18$, $LR^{-} = 0.1053$ SVD of A, energy 90% (Task 2) Hessian, κ, w*, GD and momentum behaviour (Tasks 3, 6, 7) Bias–variance decomposition of the score model (Task 8)

Why sensitivity and specificity stay fixed while the posteriors change: sensitivity P(+ | R) and specificity P(− | ¬R) are conditional on the true state, i.e. properties of the test itself. The posteriors condition on the test result and, through Bayes' rule (posterior odds = LR × prior odds), include the prior, i.e. the prevalence. Changing the prior from 10% to 2% leaves the likelihoods unchanged but divides the prior odds from 0.1111 to 0.0204.

B. Re-examination (2 marks)

$$P(+) = 0.9 \cdot 0.02 + 0.05 \cdot 0.98 = 0.018 + 0.049 = 0.0670$$

$$P(R \mid +) = 0.018 / 0.067 = 0.2687 \qquad P(R \mid -) = 0.002 / (0.002 + 0.931) = 0.002 / 0.933 = 0.0021$$

(Figure 3, right.) Odds check: $18 \times 0.0204 = 0.3673 \Rightarrow 0.3673/1.3673 = 0.2687$ ✓.

Per 1,000 patients	Prevalence 10%	Prevalence 2%	Change
Referable patients	100	20	−80
True positives	90	18	−72
False negatives (missed cases)	10	2	−8
False positives (false referrals)	45	49	+4
Total referred	135	67	−68
Share of referrals that are false	33.3%	73.1%	×2.2
P(R | +)	0.6667	0.2687	−0.3980
P(R | −)	0.0116	0.0021	−0.0095

Missed cases fall because there are far fewer diseased patients, but false referrals do not (49 versus 45): nearly three out of four referrals are now unnecessary (49 false versus 18 true, 2.72 : 1).

C. Analysis of policy options (2 marks)

Minimum specificity for P(R | +) ≥ 0.5 at 2% prevalence (sensitivity kept at 0.90):

$$0.9 \cdot 0.02 / (0.9 \cdot 0.02 + (1 - \mathrm{Sp}) \cdot 0.98) \geq 0.5 \quad \Leftrightarrow \quad 0.018 \geq (1 - \mathrm{Sp}) \cdot 0.98 \quad \Leftrightarrow \quad 1 - \mathrm{Sp} \leq 0.018367$$

$$\mathrm{Sp} \geq 1 - 0.9 \cdot 0.02 / 0.98 = 0.9816 \quad \text{(versus the present 0.9500)}$$

Moving the decision threshold. Raising the threshold lowers false positives (specificity ↑) but also lowers sensitivity; the relation is the ROC curve, which is not given. To quantify the trade-off I assume an equal-variance binormal ROC calibrated to the stated operating point ($d' = z_{0.90} + z_{0.95} = 1.2816 + 1.6449 = 2.9264$); this is an illustrative model, not measured data. Under it, the cost per 1,000 patients is $C = 10 \cdot FN + 1 \cdot FP$ (Figure 6).

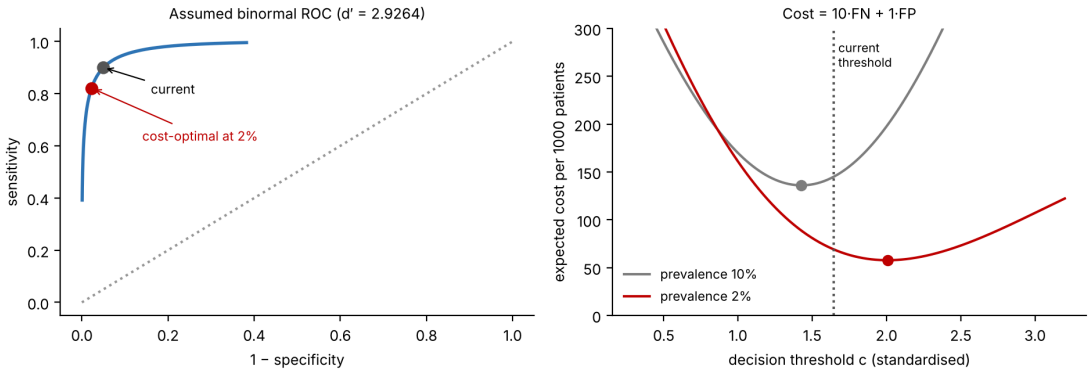


Figure 6: assumed binormal ROC with the current and the cost-optimal operating points (left); expected cost versus threshold at 10% and 2% prevalence (right).

Option (per 1,000 patients, prevalence 2%)	Sens.	Spec.	FN	FP	P(R|+)	Cost 10·FN + FP
1. Retain current threshold	0.9000	0.9500	2.00	49.00	0.2687	69.00
2. Raise threshold until P(R|+) = 0.5 on the ROC (c = 2.145)	0.7827	0.9840	4.35	15.65	0.5000	59.11
3. Cost-optimal threshold (c = 2.006; refer if LR ≥ 4.9)	0.8212	0.9776	3.58	21.97	0.4278	57.72
4. Lower threshold for more sensitivity	0.9500	0.9000	1.00	98.00	0.1624	108.00
5. Redesign: second independent confirmatory read (positive on both)†	0.8100	0.9975	3.80	2.45	0.8686	40.45
Reference: refer everybody / refer nobody	1 / 0	0 / 1	0 / 20	980 / 0	–	980 / 200
Reference: D6-level performance, no change	0.8000	0.9000	4.00	98.00	0.1404	138.00

† Assumes the second read is independent of the first with the same sensitivity and specificity (e.g. a human grader or a different imaging modality); errors of the same model are correlated, so the real gain would be smaller. The cost of the second read itself is not included.

- **Decision rule:** under the 10 : 1 cost ratio the optimal action is to refer when $P(R \mid x) \geq 1/11 = 0.0909$, i.e. when the likelihood ratio exceeds $(1 - \pi)/(10\pi) = 0.98/0.20 = 4.9$. At 10% prevalence the corresponding cut-off is LR ≥ 0.9, so the best threshold at 2% is **higher** (stricter) than at 10%.
- **Cost comparison:** the present threshold costs 69.00 units per 1,000 at 2% prevalence (145.00 at 10%). Option 3 reduces it to 57.72 (−16.4%) and cuts referrals from 67 to 38 per 1,000, at the price of 1.58 extra missed cases per 1,000. Raising specificity to 0.9816 while keeping sensitivity at 0.90 (cost 38.0) is not available on the assumed ROC: it would need $d' = 3.370$ (AUC 0.9914 instead of 0.9807), i.e. a better model.
- **Note:** $P(R \mid +) \geq 0.5$ is a workload target, not the cost-optimal target; with a 10 : 1 cost ratio the cost-optimal policy accepts a lower $P(R \mid +)$ (0.4278) in order to keep more cases.

D. Evidence-based conclusion (2 marks)

Decision: MODIFY the screening policy (keep the model; change the operating point and workflow). A full redesign is not needed unless local validation shows lower performance.

- **Recomputed probabilities and costs:** at 2% prevalence $P(R \mid +)$ falls from 0.6667 to 0.2687 and 73.1% of referrals are unnecessary, so the old threshold is no longer appropriate. The cost-optimal threshold lowers the expected cost from 69.00 to 57.72 per 1,000 and the referrals from 67 to 38. Retaining the policy unchanged would waste referrals; it is also not cost-optimal.

- **Cost of missed cases:** a missed case is priced at 10 units, and this is what keeps the sensitivity from being lowered further – lowering it to 0.78 to reach $P(R \mid +) = 0.5$ (Option 2) costs 59.11, more than Option 3. $P(R \mid -) = 0.0021$ (about 2 missed per 1,000) stays small, but the cost-optimal threshold raises misses from 2.00 to 3.58 per 1,000, which the clinical team must explicitly accept.
- **Practical rural constraints:** a false referral means travel to a district hospital and uses scarce specialist time. Recommended workflow: (i) screen with the recalibrated threshold (refer if calibrated $P(R \mid x) \geq 0.0909$); (ii) have a local tele-grader confirm positives before referral – even a partly effective second read moves the false-referral count towards the 2.45 per 1,000 of Option 5; (iii) re-measure sensitivity and specificity in the district (about 2,700 patients, or 54 referable cases, give ±0.08 at 95% confidence for sensitivity at 2% prevalence).
- **Trigger for redesign:** D6 measured recall 0.80 and specificity 0.90. If local validation confirms them, $P(R \mid +)$ at 2% falls to 0.1404 and the cost rises to 138.00 per 1,000 (twice the claimed 69.00): the model itself would then have to be redesigned (more or better training data, class-weighted cross-entropy with weight 10 on referable cases, better discrimination, $d' \geq 3.37$).